

7.3 Dot Product

Introduction In this and the following section, we shall consider two kinds of products between vectors that originated in the study of mechanics and electricity and magnetism. The first of these products, known as the **dot product**, is studied in this section.

Component Form of the Dot Product The dot product, defined next, is also known as the **inner product**, or **scalar product**. The dot product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is denoted by $\mathbf{a} \cdot \mathbf{b}$ and is a real number, or scalar, defined in terms of the components of the vectors.

Definition 7.3.1 Dot Product of Two Vectors

In 2-space the **dot product** of two vectors $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$ is the number

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2. \quad (1)$$

In 3-space the **dot product** of two vectors $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ is the number

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3. \quad (2)$$

EXAMPLE 1 Dot Product Using (2)

If $\mathbf{a} = 10\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$ and $\mathbf{b} = -\frac{1}{2}\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$, then it follows from (2) that

$$\mathbf{a} \cdot \mathbf{b} = (10)(-\frac{1}{2}) + (2)(4) + (-6)(-3) = 21.$$

EXAMPLE 2 Dot Products of the Basis Vectors

Since $\mathbf{i} = \langle 1, 0, 0 \rangle$, $\mathbf{j} = \langle 0, 1, 0 \rangle$, and $\mathbf{k} = \langle 0, 0, 1 \rangle$, we see from (2) that

$$\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{i} = 0, \quad \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{j} = 0, \quad \text{and} \quad \mathbf{k} \cdot \mathbf{i} = \mathbf{i} \cdot \mathbf{k} = 0. \quad (3)$$

Similarly, by (2)

$$\mathbf{i} \cdot \mathbf{i} = 1, \quad \mathbf{j} \cdot \mathbf{j} = 1, \quad \text{and} \quad \mathbf{k} \cdot \mathbf{k} = 1. \quad (4)$$

Properties The dot product possesses the following properties.

Theorem 7.3.1 Properties of the Dot Product

- (i) $\mathbf{a} \cdot \mathbf{b} = 0$ if $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$
- (ii) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ ← commutative law
- (iii) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ ← distributive law
- (iv) $\mathbf{a} \cdot (k\mathbf{b}) = (k\mathbf{a}) \cdot \mathbf{b} = k(\mathbf{a} \cdot \mathbf{b})$, k a scalar
- (v) $\mathbf{a} \cdot \mathbf{a} \geq 0$
- (vi) $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$

PROOF: All the properties can be proved directly from (2). We illustrate by proving parts (iii) and (vi).

To prove part (iii) we let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$. Then

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= \langle a_1, a_2, a_3 \rangle \cdot (\langle b_1, b_2, b_3 \rangle + \langle c_1, c_2, c_3 \rangle) \\ &= \langle a_1, a_2, a_3 \rangle \cdot \langle b_1 + c_1, b_2 + c_2, b_3 + c_3 \rangle \\ &= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) \\ &= a_1b_1 + a_1c_1 + a_2b_2 + a_2c_2 + a_3b_3 + a_3c_3 \quad \leftarrow \begin{cases} \text{since multiplication of real} \\ \text{numbers is distributive over} \\ \text{addition} \end{cases} \\ &= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3) \\ &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}. \end{aligned}$$

To prove part (vi) we note that

$$\mathbf{a} \cdot \mathbf{a} = \langle a_1, a_2, a_3 \rangle \cdot \langle a_1, a_2, a_3 \rangle = a_1^2 + a_2^2 + a_3^2 = \|\mathbf{a}\|^2. \quad \equiv$$

Notice that (vi) of Theorem 7.3.1 states that the **magnitude** of a vector can be written in terms of the dot product:

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

Alternative Form The dot product of two vectors $\mathbf{a}$ and $\mathbf{b}$ can also be expressed in terms of the lengths of the vectors and the angle between them. If the vectors $\mathbf{a}$ and $\mathbf{b}$ are positioned in such a manner that their initial points coincide, then we define the angle between $\mathbf{a}$ and $\mathbf{b}$ as the angle θ that satisfies $0 \leq \theta \leq \pi$.

Theorem 7.3.2 Alternative Form of Dot Product

The dot product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta \quad (5)$$

where θ is the angle between the vectors.

This more geometric form is what is generally used as the definition of the dot product in a physics course.

PROOF: Suppose θ is the angle between the vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$. Then the vector

$$\mathbf{c} = \mathbf{b} - \mathbf{a} = (b_1 - a_1)\mathbf{i} + (b_2 - a_2)\mathbf{j} + (b_3 - a_3)\mathbf{k}$$

is the third side of the triangle indicated in **FIGURE 7.3.1**. By the law of cosines we can write $\|\mathbf{c}\|^2 = \|\mathbf{b}\|^2 + \|\mathbf{a}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$ or

$$\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta = \frac{1}{2} (\|\mathbf{b}\|^2 + \|\mathbf{a}\|^2 - \|\mathbf{c}\|^2). \quad (6)$$

Using

$$\|\mathbf{a}\|^2 = a_1^2 + a_2^2 + a_3^2, \quad \|\mathbf{b}\|^2 = b_1^2 + b_2^2 + b_3^2,$$

and

$$\|\mathbf{c}\|^2 = \|\mathbf{b} - \mathbf{a}\|^2 = (b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2,$$

the right-hand side of equation (6) simplifies to $a_1b_1 + a_2b_2 + a_3b_3$. Since the last expression is (2) in Definition 7.3.1, we see that $\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta = \mathbf{a} \cdot \mathbf{b}$. $\equiv$

Orthogonal Vectors If $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors, then $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$ are both positive numbers. In this case it follows from (5) that the sign of the dot product $\mathbf{a} \cdot \mathbf{b}$ is the same as the sign of $\cos \theta$. In **FIGURE 7.3.2** we see various orientations of two vectors for angles that satisfy $0 \leq \theta \leq \pi$. Observe from Figure 7.3.2 and (5) that $\mathbf{a} \cdot \mathbf{b} > 0$ for $\theta = 0$, $\mathbf{a} \cdot \mathbf{b} > 0$ when θ is acute, $\mathbf{a} \cdot \mathbf{b} = 0$ for $\theta = \pi/2$, $\mathbf{a} \cdot \mathbf{b} < 0$ when θ is obtuse, and $\mathbf{a} \cdot \mathbf{b} < 0$ for $\theta = \pi$. In the special case $\theta = \pi/2$ we say that the vectors are **orthogonal** or **perpendicular**. Moreover, if we know that $\mathbf{a} \cdot \mathbf{b} = 0$, then the only angle for which this is true and satisfying $0 \leq \theta \leq \pi$ is $\theta = \pi/2$. This leads us to the next result.

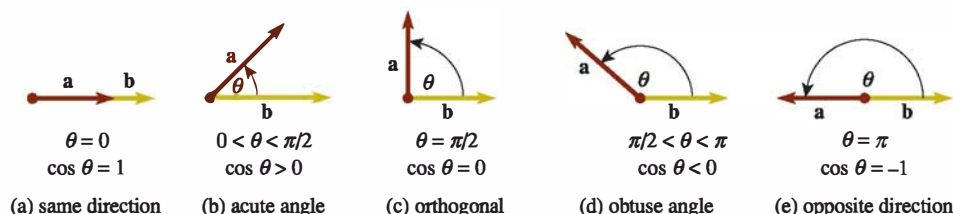


FIGURE 7.3.2 Angles between vectors $\mathbf{a}$ and $\mathbf{b}$

Theorem 7.3.3 Criterion for Orthogonal Vectors

Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.

Since $\mathbf{0} \cdot \mathbf{b} = \langle 0, 0, 0 \rangle \cdot \langle b_1, b_2, b_3 \rangle = 0(b_1) + 0(b_2) + 0(b_3) = 0$ for every vector $\mathbf{b}$, the zero vector $\mathbf{0}$ is considered to be orthogonal to every vector.

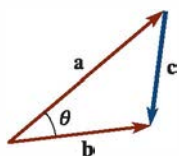


FIGURE 7.3.1 Vector $\mathbf{c}$ in the proof of Theorem 7.3.2

EXAMPLE 3 Orthogonal Vectors

If $\mathbf{a} = -3\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} + 14\mathbf{j} + 5\mathbf{k}$, then

$$\mathbf{a} \cdot \mathbf{b} = (-3)(2) + (-1)(14) + (4)(5) = 0.$$

From Theorem 7.3.3, we conclude that $\mathbf{a}$ and $\mathbf{b}$ are orthogonal. $\equiv$

In Example 1, from $\mathbf{a} \cdot \mathbf{b} \neq 0$ we can conclude that the vectors $\mathbf{a} = 10\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$ and $\mathbf{b} = -\frac{1}{2}\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ are *not* orthogonal. From (3) of Example 2, we see what is apparent in Figure 7.2.8, namely, the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ constitute a set of mutually orthogonal unit vectors.

Angle Between Two Vectors By equating the two forms of the dot product, (2) and (5), we can determine the angle between two vectors from

$$\cos \theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\|\mathbf{a}\| \|\mathbf{b}\|}. \quad (7)$$

EXAMPLE 4 Angle Between Two Vectors

Find the angle between $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 5\mathbf{j} + \mathbf{k}$.

SOLUTION From $\|\mathbf{a}\| = \sqrt{14}$, $\|\mathbf{b}\| = \sqrt{27}$, $\mathbf{a} \cdot \mathbf{b} = 14$, we see from (7) that

$$\cos \theta = \frac{14}{\sqrt{14}\sqrt{27}} = \frac{\sqrt{42}}{9}$$

and so $\theta = \cos^{-1}(\sqrt{42}/9) \approx 0.77$ radian or $\theta \approx 43.9^\circ$. $\equiv$

Direction Cosines For a nonzero vector $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ in 3-space, the angles α , β , and γ between $\mathbf{a}$ and the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, respectively, are called **direction angles** of $\mathbf{a}$. See **FIGURE 7.3.3**. Now, by (7),

$$\cos \alpha = \frac{\mathbf{a} \cdot \mathbf{i}}{\|\mathbf{a}\| \|\mathbf{i}\|}, \quad \cos \beta = \frac{\mathbf{a} \cdot \mathbf{j}}{\|\mathbf{a}\| \|\mathbf{j}\|}, \quad \cos \gamma = \frac{\mathbf{a} \cdot \mathbf{k}}{\|\mathbf{a}\| \|\mathbf{k}\|},$$

which simplify to

$$\cos \alpha = \frac{a_1}{\|\mathbf{a}\|}, \quad \cos \beta = \frac{a_2}{\|\mathbf{a}\|}, \quad \cos \gamma = \frac{a_3}{\|\mathbf{a}\|}.$$

We say that $\cos \alpha$, $\cos \beta$, and $\cos \gamma$ are the **direction cosines** of $\mathbf{a}$. The direction cosines of a nonzero vector $\mathbf{a}$ are simply the components of the unit vector $(1/\|\mathbf{a}\|)\mathbf{a}$:

$$\frac{1}{\|\mathbf{a}\|} \mathbf{a} = \frac{a_1}{\|\mathbf{a}\|} \mathbf{i} + \frac{a_2}{\|\mathbf{a}\|} \mathbf{j} + \frac{a_3}{\|\mathbf{a}\|} \mathbf{k} = (\cos \alpha) \mathbf{i} + (\cos \beta) \mathbf{j} + (\cos \gamma) \mathbf{k}.$$

Since the magnitude of $(1/\|\mathbf{a}\|)\mathbf{a}$ is 1, it follows from the last equation that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.$$

EXAMPLE 5 Direction Cosines/Angles

Find the direction cosines and direction angles of the vector $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$.

SOLUTION From $\|\mathbf{a}\| = \sqrt{2^2 + 5^2 + 4^2} = \sqrt{45} = 3\sqrt{5}$, we see that the direction cosines are

$$\cos \alpha = \frac{2}{3\sqrt{5}}, \quad \cos \beta = \frac{5}{3\sqrt{5}}, \quad \cos \gamma = \frac{4}{3\sqrt{5}}.$$

The direction angles are

$$\alpha = \cos^{-1}\left(\frac{2}{3\sqrt{5}}\right) \approx 1.27 \text{ radians} \quad \text{or} \quad \alpha \approx 72.7^\circ$$

$$\beta = \cos^{-1}\left(\frac{5}{3\sqrt{5}}\right) \approx 0.73 \text{ radian} \quad \text{or} \quad \beta \approx 41.8^\circ$$

$$\gamma = \cos^{-1}\left(\frac{4}{3\sqrt{5}}\right) \approx 0.93 \text{ radian} \quad \text{or} \quad \gamma \approx 53.4^\circ. \quad \equiv$$

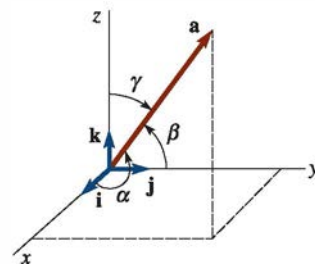


FIGURE 7.3.3 Direction angles α , β , and γ

Observe in Example 5 that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{4}{45} + \frac{25}{45} + \frac{16}{45} = 1.$$

Component of $\mathbf{a}$ on $\mathbf{b}$ The distributive law and (3) enable us to express the components of a vector $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ in terms of the dot product:

$$a_1 = \mathbf{a} \cdot \mathbf{i}, \quad a_2 = \mathbf{a} \cdot \mathbf{j}, \quad a_3 = \mathbf{a} \cdot \mathbf{k}. \quad (8)$$

Symbolically, we write the components of $\mathbf{a}$ as

$$\text{comp}_{\mathbf{i}}\mathbf{a} = \mathbf{a} \cdot \mathbf{i}, \quad \text{comp}_{\mathbf{j}}\mathbf{a} = \mathbf{a} \cdot \mathbf{j}, \quad \text{comp}_{\mathbf{k}}\mathbf{a} = \mathbf{a} \cdot \mathbf{k}. \quad (9)$$

We shall now see that the results indicated in (9) carry over to finding the **component of $\mathbf{a}$ on an arbitrary vector $\mathbf{b}$** . Note that in either of the two cases shown in **FIGURE 7.3.4**,

$$\text{comp}_{\mathbf{b}}\mathbf{a} = \|\mathbf{a}\| \cos \theta. \quad (10)$$

In Figure 7.3.4(b), $\text{comp}_{\mathbf{b}}\mathbf{a} < 0$ since $\pi/2 < \theta \leq \pi$. Now, by writing (10) as

$$\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta}{\|\mathbf{b}\|} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|},$$

we see that

$$\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a} \cdot \left(\frac{1}{\|\mathbf{b}\|} \mathbf{b} \right) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|}. \quad (11)$$

In other words:

To find the component of $\mathbf{a}$ on a vector $\mathbf{b}$, we dot $\mathbf{a}$ with a unit vector in the direction of $\mathbf{b}$.

EXAMPLE 6 Component of a Vector on Another Vector

Let $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$. Find $\text{comp}_{\mathbf{b}}\mathbf{a}$ and $\text{comp}_{\mathbf{a}}\mathbf{b}$.

SOLUTION We first form a unit vector in the direction of $\mathbf{b}$:

$$\|\mathbf{b}\| = \sqrt{6}, \quad \frac{1}{\|\mathbf{b}\|} \mathbf{b} = \frac{1}{\sqrt{6}} (\mathbf{i} + \mathbf{j} + 2\mathbf{k}).$$

Then from (11) we have

$$\text{comp}_{\mathbf{b}}\mathbf{a} = (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}) \cdot \frac{1}{\sqrt{6}} (\mathbf{i} + \mathbf{j} + 2\mathbf{k}) = -\frac{3}{\sqrt{6}}.$$

By modifying (11) accordingly, we have

$$\text{comp}_{\mathbf{a}}\mathbf{b} = \mathbf{b} \cdot \left(\frac{1}{\|\mathbf{a}\|} \mathbf{a} \right).$$

$$\text{Therefore,} \quad \|\mathbf{a}\| = \sqrt{29}, \quad \frac{1}{\|\mathbf{a}\|} \mathbf{a} = \frac{1}{\sqrt{29}} (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k})$$

$$\text{and} \quad \text{comp}_{\mathbf{a}}\mathbf{b} = (\mathbf{i} + \mathbf{j} + 2\mathbf{k}) \cdot \frac{1}{\sqrt{29}} (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}) = -\frac{3}{\sqrt{29}}. \quad \equiv$$

Projection of $\mathbf{a}$ onto $\mathbf{b}$ As illustrated in **FIGURE 7.3.5**, the **projection** of a vector $\mathbf{a}$ in any of the directions determined by $\mathbf{i}, \mathbf{j}, \mathbf{k}$ is simply the **vector** formed by multiplying the component of $\mathbf{a}$ in the specified direction with a unit vector in that direction; for example,

$$\text{proj}_{\mathbf{i}}\mathbf{a} = (\text{comp}_{\mathbf{i}}\mathbf{a})\mathbf{i} = (\mathbf{a} \cdot \mathbf{i})\mathbf{i} = a_1\mathbf{i}$$

and so on. **FIGURE 7.3.6** shows the general case of the **projection of $\mathbf{a}$ onto $\mathbf{b}$** :

$$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a}) \left(\frac{1}{\|\mathbf{b}\|} \mathbf{b} \right) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b}. \quad (12)$$

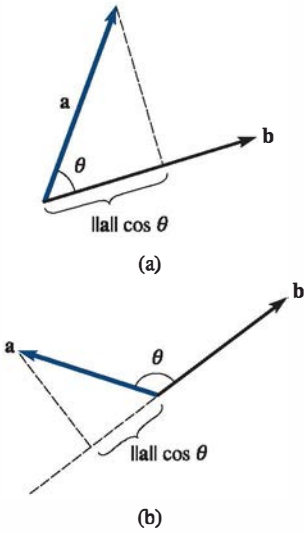


FIGURE 7.3.4 Component of $\mathbf{a}$ on $\mathbf{b}$

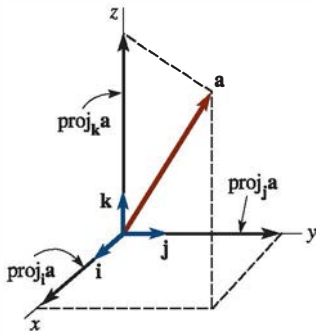


FIGURE 7.3.5 Projections of $\mathbf{a}$ onto $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$

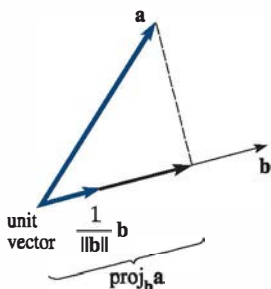


FIGURE 7.3.6 Projection of $\mathbf{a}$ onto $\mathbf{b}$

EXAMPLE 7 Projection of a Vector on Another Vector

Find the projection of $\mathbf{a} = 4\mathbf{i} + \mathbf{j}$ onto the vector $\mathbf{b} = 2\mathbf{i} + 3\mathbf{j}$. Graph.

SOLUTION First, we find the component of $\mathbf{a}$ and $\mathbf{b}$. Since $\|\mathbf{b}\| = \sqrt{13}$, we find from (11) that

$$\text{comp}_{\mathbf{b}}\mathbf{a} = (4\mathbf{i} + \mathbf{j}) \cdot \frac{1}{\sqrt{13}}(2\mathbf{i} + 3\mathbf{j}) = \frac{11}{\sqrt{13}}.$$

Thus, from (12),

$$\text{proj}_{\mathbf{b}}\mathbf{a} = \left(\frac{11}{\sqrt{13}}\right)\left(\frac{1}{\sqrt{13}}\right)(2\mathbf{i} + 3\mathbf{j}) = \frac{22}{13}\mathbf{i} + \frac{33}{13}\mathbf{j}.$$

The graph of this vector is shown in **FIGURE 7.3.7**.

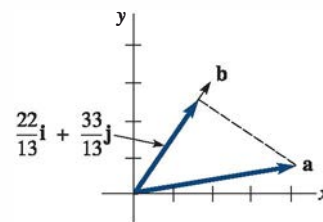


FIGURE 7.3.7 Projection of $\mathbf{a}$ onto $\mathbf{b}$ in Example 7

Physical Interpretation of the Dot Product When a constant force of magnitude F moves an object a distance d in the same direction of the force, the work done is simply $W = Fd$. However, if a constant force $\mathbf{F}$ applied to a body acts at an angle θ to the direction of motion, then the work done by $\mathbf{F}$ is defined to be the product of the component of $\mathbf{F}$ in the direction of the displacement and the distance $\|\mathbf{d}\|$ that the body moves:

$$W = (\|\mathbf{F}\| \cos \theta) \|\mathbf{d}\| = \|\mathbf{F}\| \|\mathbf{d}\| \cos \theta.$$

See **FIGURE 7.3.8**. It follows from Theorem 7.3.2 that if $\mathbf{F}$ causes a displacement $\mathbf{d}$ of a body, then the work done is

$$W = \mathbf{F} \cdot \mathbf{d}. \quad (13)$$

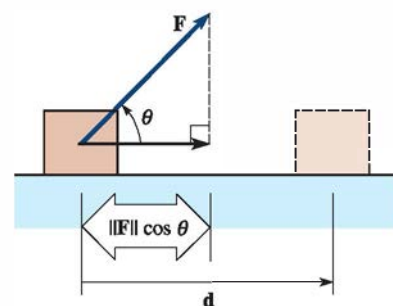


FIGURE 7.3.8 Work done by a force $\mathbf{F}$

EXAMPLE 8 Work Done by a Constant Force

Find the work done by a constant force $\mathbf{F} = 2\mathbf{i} + 4\mathbf{j}$ if its point of application to a block moves from $P_1(1, 1)$ to $P_2(4, 6)$. Assume that $\|\mathbf{F}\|$ is measured in newtons and $\|\mathbf{d}\|$ is measured in meters.

SOLUTION The displacement of the block is given by

$$\mathbf{d} = \overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = 3\mathbf{i} + 5\mathbf{j}.$$

It follows from (13) that the work done is

$$W = (2\mathbf{i} + 4\mathbf{j}) \cdot (3\mathbf{i} + 5\mathbf{j}) = 26 \text{ N}\cdot\text{m}.$$

7.3 Exercises

Answers to selected odd-numbered problems begin on page ANS-15.

In Problems 1–12, $\mathbf{a} = \langle 2, -3, 4 \rangle$, $\mathbf{b} = \langle -1, 2, 5 \rangle$, and $\mathbf{c} = \langle 3, 6, -1 \rangle$. Find the indicated scalar or vector.

1. $\mathbf{a} \cdot \mathbf{b}$
2. $\mathbf{b} \cdot \mathbf{c}$
3. $\mathbf{a} \cdot \mathbf{c}$
4. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$
5. $\mathbf{a} \cdot (4\mathbf{b})$
6. $\mathbf{b} \cdot (\mathbf{a} - \mathbf{c})$
7. $\mathbf{a} \cdot \mathbf{a}$
8. $(2\mathbf{b}) \cdot (3\mathbf{c})$
9. $\mathbf{a} \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c})$
10. $(2\mathbf{a}) \cdot (\mathbf{a} - 2\mathbf{b})$
11. $\left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}}\right) \mathbf{b}$
12. $(\mathbf{c} \cdot \mathbf{b}) \mathbf{a}$

In Problems 13 and 14, find $\mathbf{a} \cdot \mathbf{b}$ if the smaller angle between $\mathbf{a}$ and $\mathbf{b}$ is as given.

13. $\|\mathbf{a}\| = 10$, $\|\mathbf{b}\| = 5$, $\theta = \pi/4$
14. $\|\mathbf{a}\| = 6$, $\|\mathbf{b}\| = 12$, $\theta = \pi/6$

15. Determine which pairs of the following vectors are orthogonal:
 - (a) $\langle 2, 0, 1 \rangle$
 - (b) $3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
 - (c) $2\mathbf{i} - \mathbf{j} - \mathbf{k}$
 - (d) $\mathbf{i} - 4\mathbf{j} + 6\mathbf{k}$
 - (e) $\langle 1, -1, 1 \rangle$
 - (f) $\langle -4, 3, 8 \rangle$
16. Determine a scalar c so that the given vectors are orthogonal.
 - (a) $\mathbf{a} = 2\mathbf{i} - c\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$
 - (b) $\mathbf{a} = \langle c, \frac{1}{2}, c \rangle$, $\mathbf{b} = \langle -3, 4, c \rangle$
17. Find a vector $\mathbf{v} = \langle x_1, y_1, 1 \rangle$ that is orthogonal to both $\mathbf{a} = \langle 3, 1, -1 \rangle$ and $\mathbf{b} = \langle -3, 2, 2 \rangle$.
18. A rhombus is an oblique-angled parallelogram with all four sides equal. Use the dot product to show that the diagonals of a rhombus are perpendicular.